

B. Tech. Degree IV Semester Examination April 2016

CS 1403 COMPUTER ARCHITECTURE AND ORGANISATION

(2012 Scheme)

Time: 3 Hours

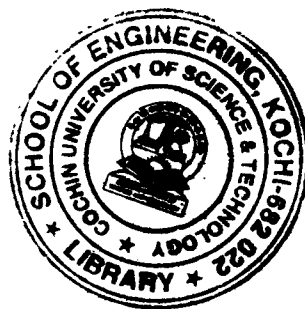
Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Write a short note on instruction sequencing.
 (b) Describe Assembler Directives with example.
 (c) Explain Booth algorithm with the help of an example.
 (d) Describe various steps in executing a complete instruction.
 (e) Explain static memory and dynamic memory.
 (f) What do you mean by memory interleaving? Explain.
 (g) Describe various bus arbitration mechanisms.
 (h) Write a short note on interrupt nesting.



PART B

(4 × 15 = 60)

- II. (a) Explain the basic operational concepts for executing a typical instruction, ADD LOC A,R0. (7)
 (b) Describe various operations to be performed while calling a subroutine. (8)

OR

- III. Define addressing mode. Describe various addressing modes in modern processors with examples. (15)

- IV. Explain the design and operation of a microprogrammed control unit in detail. (15)

OR

- V. (a) Explain the operation of a 4 bit carry-lookahead adder. (7)
 (b) Explain the fast multiplication algorithm of 'Bit' pair recoding. (8)

- VI. Explain cache memory and various cache mapping functions in detail. (15)

OR

- VII. (a) Explain in detail virtual memory and address translation. (10)
 (b) Write a short note on secondary storage devices. (5)

- VIII. (a) Explain the detailed operation of DMA. (10)
 (b) Write a note on vectored interrupts. (5)

OR

- IX. Write short notes on: (15)

- (i) Synchronous bus.
 (ii) Asynchronous bus.
 (iii) USB.